Exercise 5.6

1.(i) $-\frac{y}{x}$ (ii) $\frac{y-2x}{2y-x}$ (iii) $-\frac{(x^2+ay)}{(y^2+ax)}$ 3. $\frac{4}{3}\left(\frac{1-4x+3y}{1+4x-3y}\right)$

Exercise 5.7

1.(i) $x^{\sin x}\left[\frac{\sin x}{x}+\cos x\log x\right]$ (ii) $(\sin x)^x[x\cot x+\log(\sin x)]$

(iii) $(\sin x)^{\tan x}[1+\sec^2 x\log(\sin x)]$

(iv) $\frac{1}{2}\sqrt{\frac{(x-1)(x-2)}{(x-3)(x^2+x+1)}}\left\{\frac{1}{x-1}+\frac{1}{x-2}-\frac{1}{x-3}-\frac{2x+1}{x^2+x+1}\right\}$

Exercise 5.8

1.(i) $-\frac{1}{t^2}$ (ii) $t\cos t$ (iii) $-\tan\theta$ (iv) $\cot\frac{\theta}{2}$

2. $-\tan x$ 3. $\frac{\sin 2x}{2x}$

Exercise 5.9

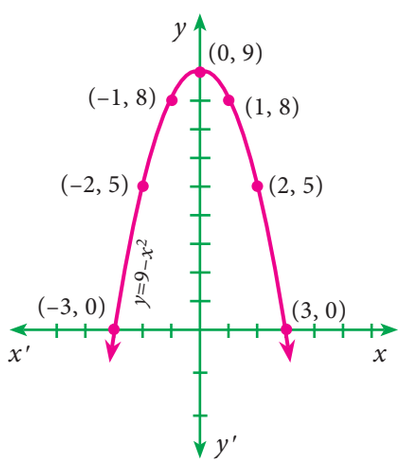
1.(i) $9y$ (ii) $-\frac{1}{x^2}+a^x(\log a)^2$ (iii) $-\frac{1}{a}\operatorname{cosec}^3\theta$

Exercise: 5.10

1	2	3	4	5	6	7	8	9	10	11	12	13
(d)	(c)	(a)	(b)	(a)	(a)	(c)	(a)	(d)	(a)	(a)	(d)	(b)
14	15	16	17	18	19	20	21	22	23	24	25	
(a)	(a)	(a)	(c)	(b)	(d)	(a)	(a)	(c)	(a)	(b)	(d)	

Miscellaneous Problems

2.



4. $-\frac{1}{10}$

6. (i) continuous at $x = 1$; differentiable at $x = 1$

(ii) not continuous at $x = 2$; not differentiable at $x = 2$

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